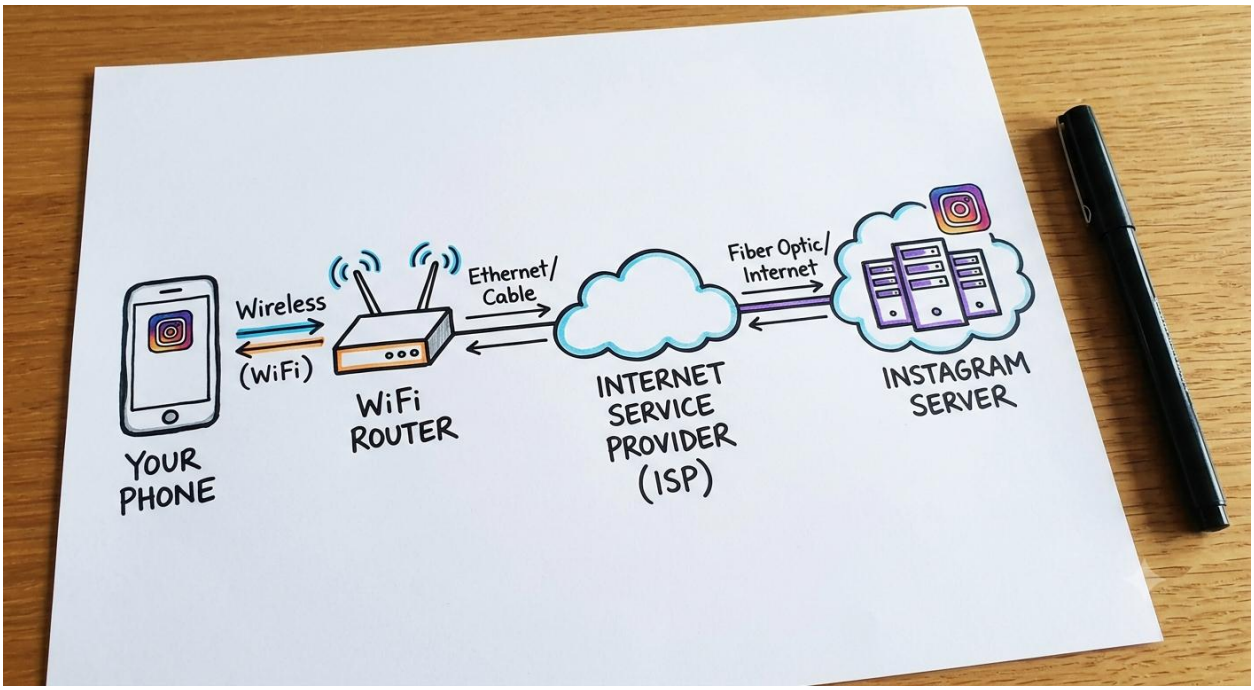
Assignment 1

Introduction of Network

1. **Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.**



2. **List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).**

No.	Device	Type of Network	Why?
1.	 Smartphone	WAN / WLAN	Connects to the internet using mobile data or Wi-Fi.
2.	 Laptop	LAN / WLAN	Connects to a local Wi-Fi network and the internet.
3.	 Smart TV	LAN / WLAN	Connects to the home Wi-Fi network to access online services.
4.	 Wi-Fi Router	LAN / WAN	Connects devices in the home LAN to the internet (WAN).
5.	 Smartwatch	PAN	Connects to a smartphone using Bluetooth, which is a Personal Area Network.

3. **Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.**

Internet	**Intranet**
A public network available worldwide.	A private network used by an organization.
Anyone can access it with an Internet connection.	Only authorized users can access it.
Example: Google, YouTube, Instagram.	Example: A college's internal student/employee portal.

- When I connect my laptop to my college/public Wi-Fi and open YouTube or Google, I am using the Internet. If the college Wi-Fi also provides access to a private college portal that only students and staff can use, that private system is an Intranet.

4. **Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app.**

Example: WhatsApp Search

- When I search for a message in **WhatsApp**, my smartphone sends the **search request through** Wi-Fi/mobile network to **WhatsApp's servers.**

- The request passes through my **router/ISP** and the Internet to **reach the server**.

- The **WhatsApp server** processes the **request** and sends the **required data** back through the **Internet.**

- My smartphone **receives the response** and **displays** the search results in the app.